



Semantic & Temporal Analysis of Social Media Text

for Student Mental Health Assistance

Project Members

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Project Overview



AI-Driven Mental Health Analysis

Using advanced NLP to analyze student social media posts for mental health indicators



Temporal Pattern Recognition

Tracking emotional trends over time to identify early warning signs and risk patterns



Explainable Insights

Transparent predictions using SHAP and LIME to understand model decisions

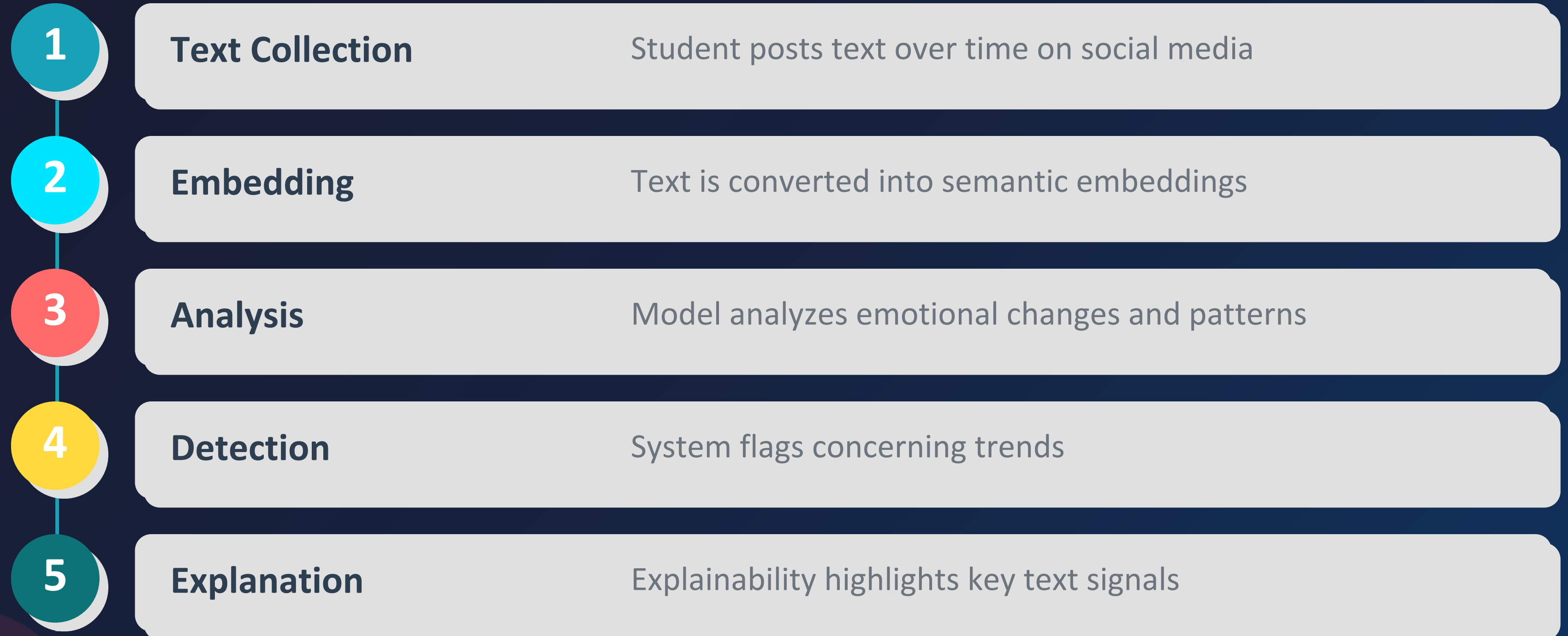
Problem Statement

CURRENT LIMITATIONS

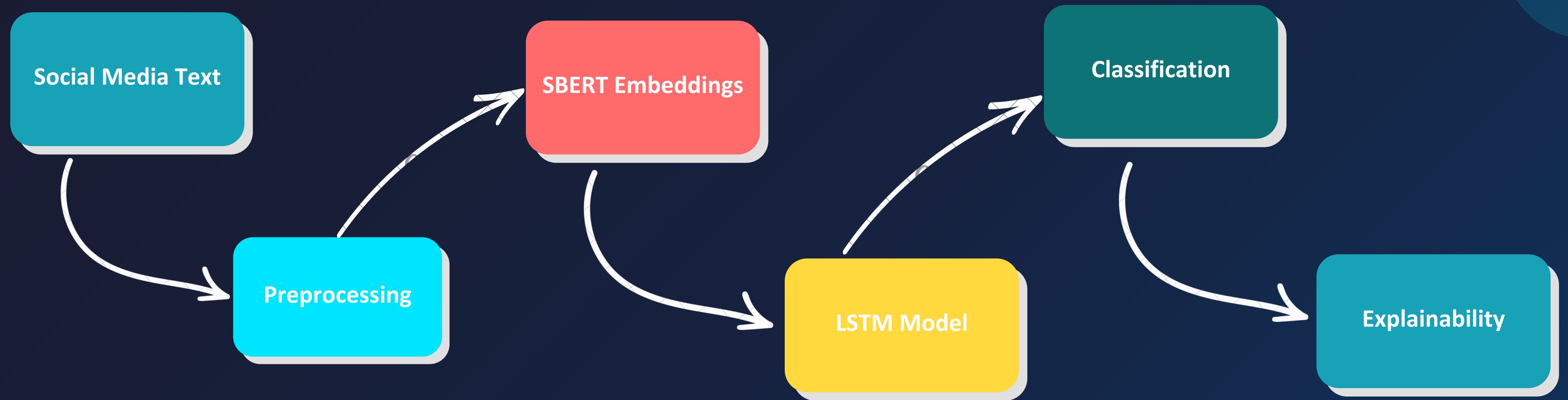
- ✗ Existing systems rely on classical ML and static text analysis
- ✗ They ignore context, sarcasm, and nuanced language
- ✗ No consideration of emotional trends over time

Example Workflow

Illustrative Process

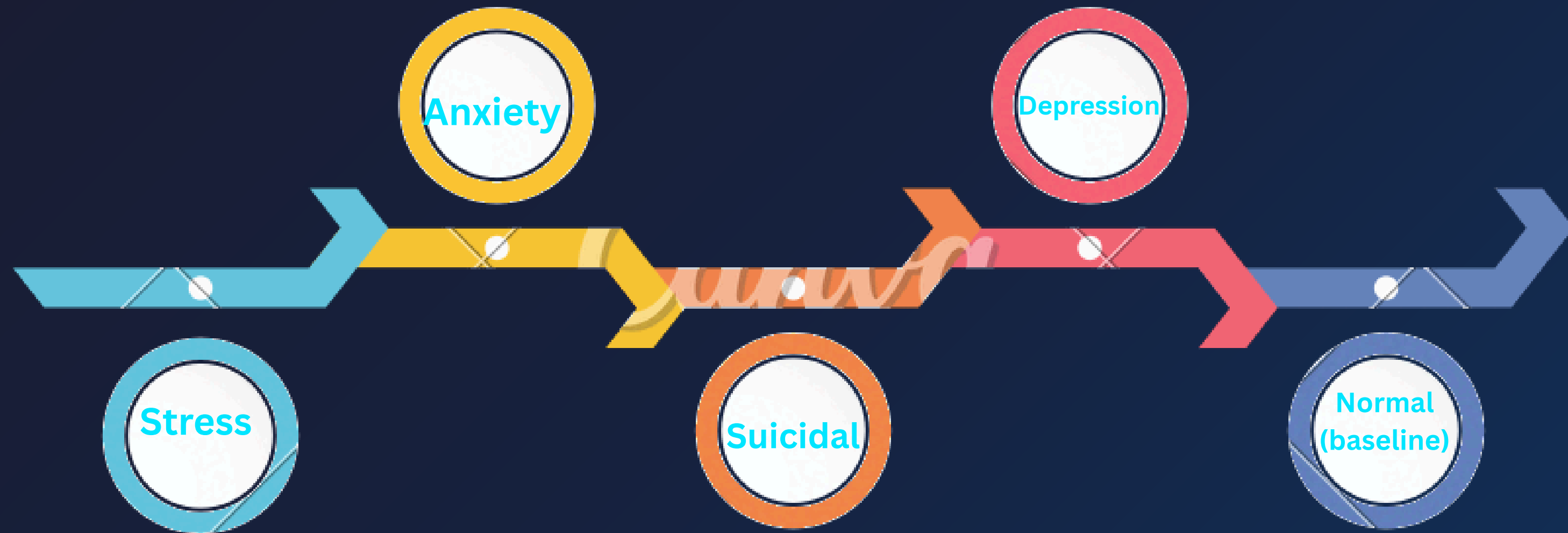


System Architecture



Dataset Overview

Mental Health Categories



Project Timeline

One Year Project • Two Semesters

M1

Data Acquisition & Preprocessing

Semester 1 (Months 1-2)

Dataset ingestion, text cleaning, normalization, and structuring. Ensures data quality and prepares student-wise chronological text data.

M2

Semantic Mental Health Classification

Semester 1 (Months 3-4)

Uses transformer models (BERT/SBERT) to extract contextual embeddings and classify text into mental health categories.

M3

Temporal Emotional Pattern Analysis

Semester 2 (Months 5-8)

Applies LSTM networks on sequences of student posts to identify emotional trends and risk levels.

M4

Explainable Mental Health Insights

Semester 2 (Months 9-12)

Integrates explainability techniques (SHAP/LIME) to provide transparent predictions and insights.

Datasets



Reddit Mental Health Dataset (Hugging Face)

<https://huggingface.co/datasets/kamruzzaman-asif/reddit-mental-health-classification>



A Reddit Dataset for Stress Analysis in Social Media

<https://huggingface.co/datasets/andreagasparini/dreaddit>
<https://aclanthology.org/D19-6213.pdf>



<https://github.com/abuDakarSiddiqurRahman/DepressionEmo/blob/main/Dataset/train.json>

Machine Learning Models



Neural Network

SoftMax activation • Binary output (0-1) classification

**SVM +
TF-IDF**
Baseline

LSTM
Sequence
Learning



Multiple Regression

Logistic regression for multi-class prediction



NLP Transformer

Deep learning for semantic understanding

**BERT /
RoBERTa**
Semantic

**SBERT +
LSTM**
Hybrid



XGBoost

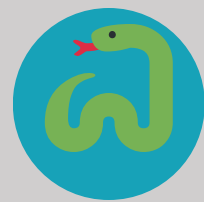
Gradient boosting for ensemble learning



SVC (Support Vector)

Support Vector Classifier for pattern recognition

Tools & Techniques



Python & Libraries

PyTorch, TensorFlow, Transformers, NLTK, scikit-learn



Data Processing & Analysis

Pandas, NumPy, Matplotlib, Seaborn for data manipulation and visualization



Development Environment

Jupyter Notebooks, Google Colab, GitHub for version control and collaboration

Planned Application Features

Four-Pillar Support System



Anxiety Management

- ✓ Breathing exercises
- ✓ Cognitive reframing tools
- ✓ Anxiety tracking timeline



Stress Relief

- ✓ Stress level assessment
- ✓ Coping strategy suggestions
- ✓ Workload balance tips



Depression Support

- ✓ Mood tracking
- ✓ Activity suggestions
- ✓ Connection resources

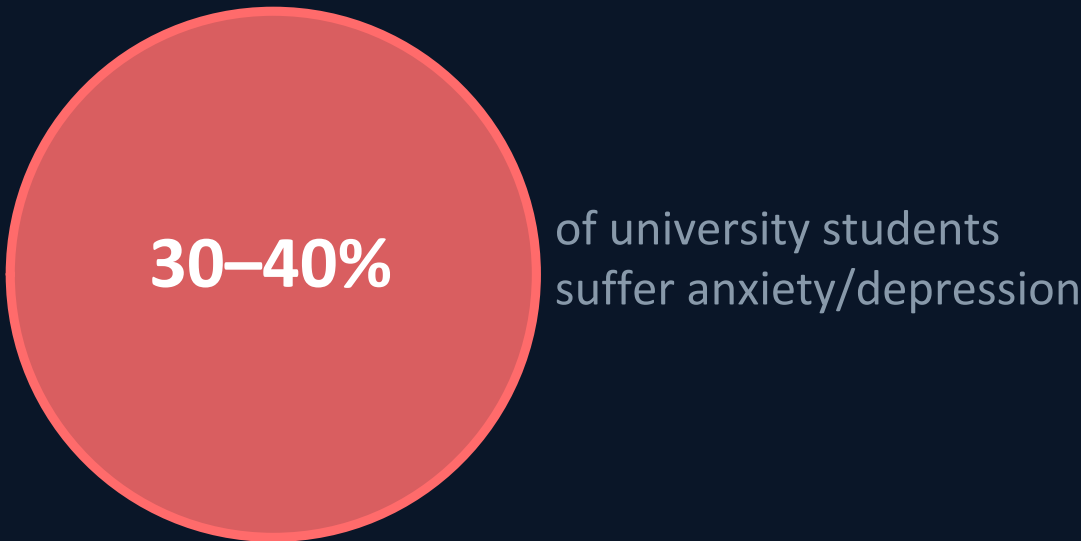


Crisis Prevention

- ✓ Risk flag detection
- ✓ Emergency resources
- ✓ Counselor connection

The Problem We're Solving

Why existing mental health assessment systems fall short



Current NLP Accuracy Ceiling:



01



Reactive, Not Proactive

Traditional surveys only run at crisis points — early-stage deterioration is completely missed and untreated.

02



No Temporal Awareness

Existing NLP tools analyze posts in isolation, ignoring gradual emotional decline patterns over 2–12 weeks.

03



Single-Modal Limitations

Relying on text alone fails — sarcasm, code-switching, and informal language create high false-negative rates.

MindCare vs. Existing Systems

How MindCare advances beyond current state-of-the-art mental health NLP tools

System	Temporal Analysis	Multi-Modal	XAI / SHAP	Counsellor Portal
Woebot (AI chatbot)	✗ No	✗ Text only	✗ No	✗ No
Koko / Crisis Text Line	✗ No	✓ Partial	✗ No	✓ Yes
MindBEAGLE (Andriot 2023)	✗ No	✗ Text only	✗ No	✗ No
SBERT+LSTM (Malhotra 2022)	✓ Yes	✗ Text only	✗ No	✗ No
Wysa (mental health app)	✗ No	✓ Partial	✗ No	✓ Limited
CLPsych shared tasks	✓ Partial	✗ Text only	✗ No	✗ No
MindCare (Ours)	✓ 2-12 wk	✓ PHQ9+Social	✓ SHAP+LIME	✓ Full

MindCare's Unique Advantages

- Only system combining LSTM temporal tracking (2-12 wks) + SBERT semantics + structured PHQ-9/GAD-7
- Full explainability via SHAP/LIME — every risk prediction is auditable, no 'black box' decisions
- 8-level validated risk framework maps to institutional escalation protocols — beyond binary detection

Implementation Approach

Four-pronged EDA — Statistical & Linguistic Analysis Pipeline

Category Distribution



- Balanced: 209,590 samples per class
- 50/50 MH vs. Control split
- Bar chart + pie chart visualisation
- Prevents model bias

Text Statistical Analysis



- MH avg: 934 chars / 177.6 words
- Control avg: 1,092 chars / 205.6 words
- Histograms, box plots, std deviation
- Reveals distributional differences

Word Frequency & N-grams



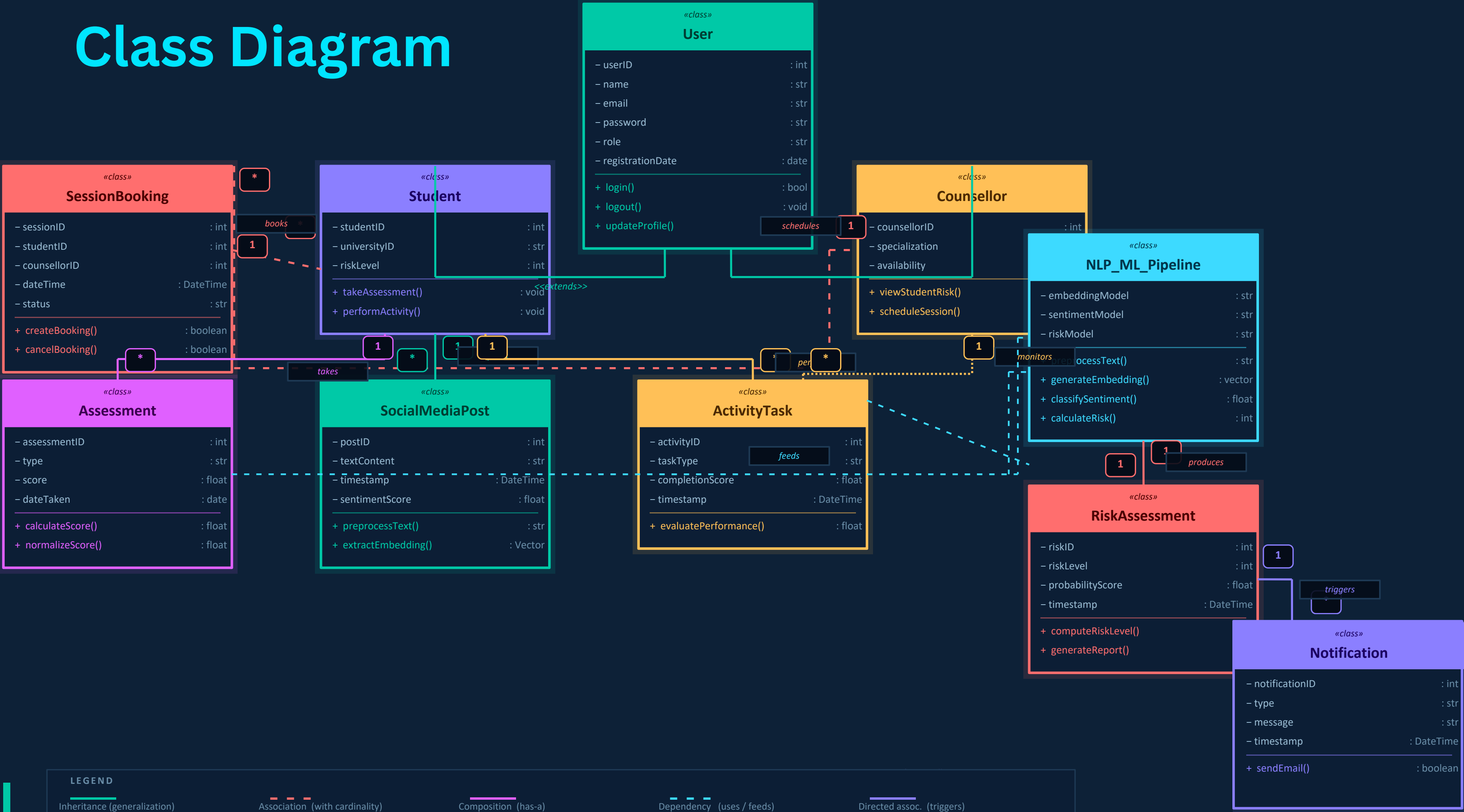
- Top 15 most frequent words per class
- MH top signals: 'feel', 'not', 'like'
- Control dominated by pronouns
- Unique vocab per class identified

Sentiment Analysis



- MH net sentiment mean = -0.26
- Control net sentiment mean = $+0.29$
- VADER-based positive/negative counts
- Confirms strong class separability

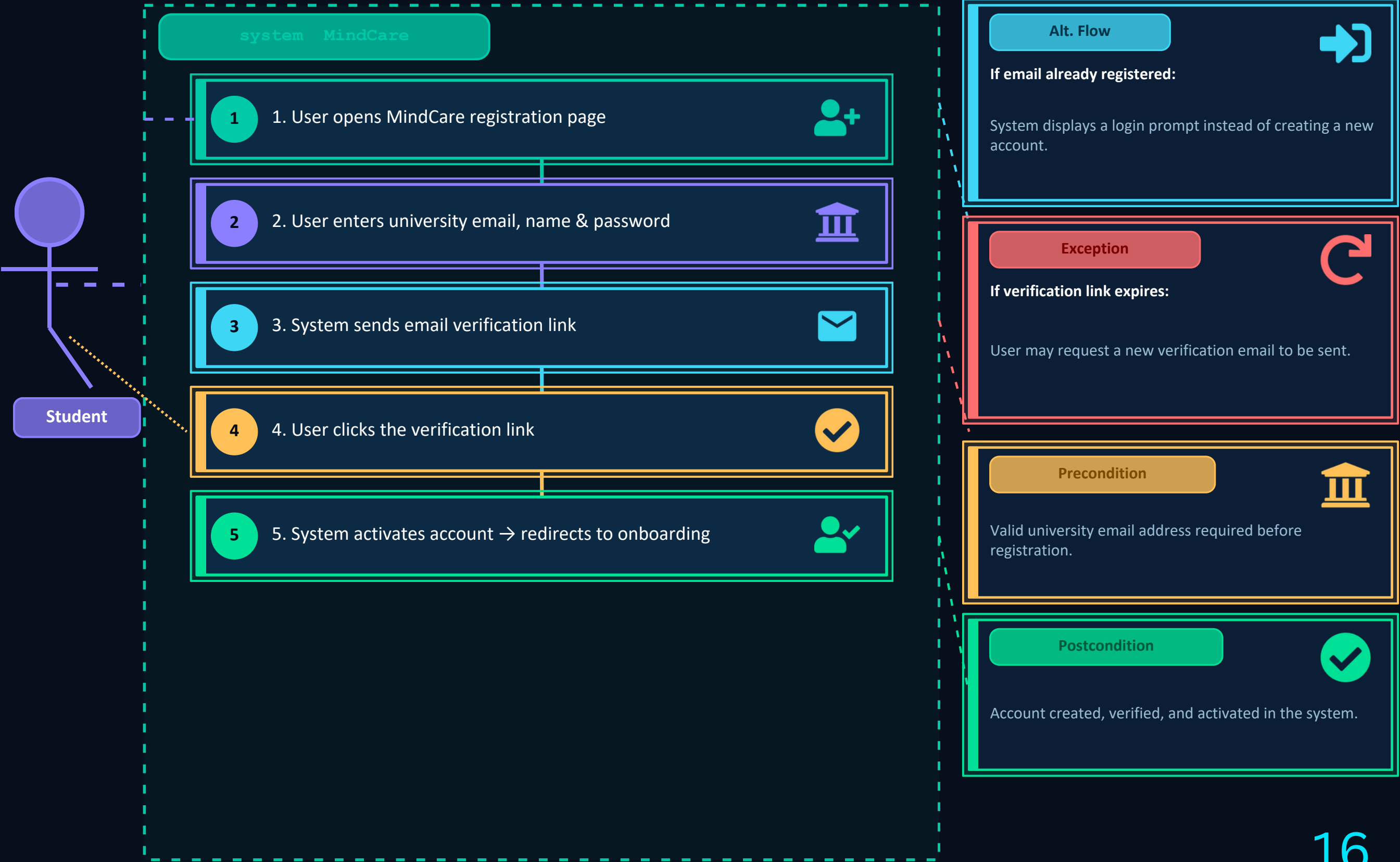
Class Diagram



Use-Case: Register Account

Actor: Student · System: MindCare

Name	Register Account
Actor	Student
Precondition	User has a valid university email address
Postcondition	Account created, verified, and activated in the system
Alt. Flow	If email is already registered, show login prompt
Exception	If verification link expires, user can request a new one
Main Flow	1 User navigates to the MindCare registration page 2 User enters full name, university email, and password 3 System sends a verification email to the provided address 4 User clicks the verification link in the email 5 System activates the account and redirects to onboarding



System Architecture — Full Pipeline

4-layer microservices: Data → NLP Pipeline → Fusion Engine → Output Layer



L1

Data Acquisition Layer

✓ DONE

Reddit dataset: 419,180 posts, 28 categories · Preprocessing: lowercasing, stop-word removal, feature extraction



L2

NLP Processing Layer

⌚ NEXT

SBERT sentence embeddings for semantic representation · VADER sentiment analysis · n-gram frequency extraction



L3

Multi-Modal Fusion Engine

⌚ NEXT

Social text (35%) + PHQ-9/GAD-7 (40%) + Activity tasks (25%) · Weighted fusion → unified risk score



L4

Output & Explainability

⌚ NEXT

8-level risk classification (Normal → Crisis 8) · SHAP/LIME explanations · Counsellor dashboard + alerts

Tech Stack

NLP

Python · Pandas · NLTK

AI/ML

HuggingFace · SBERT

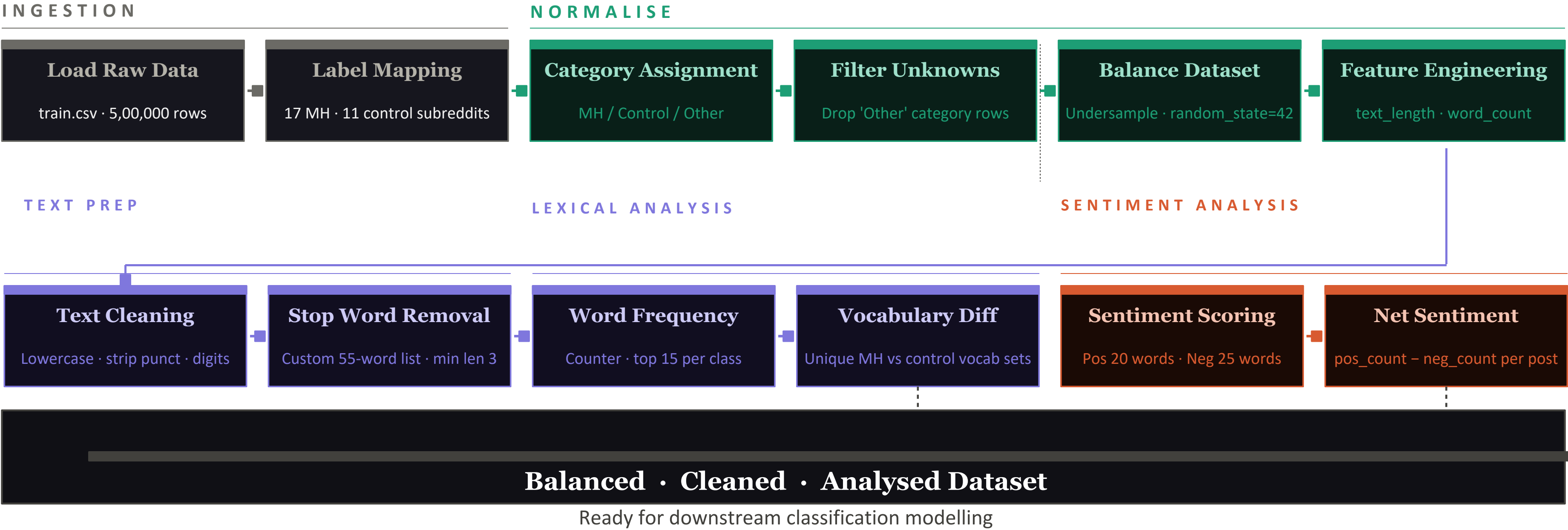
DB

PostgreSQL · MongoDB · Pinecone

App

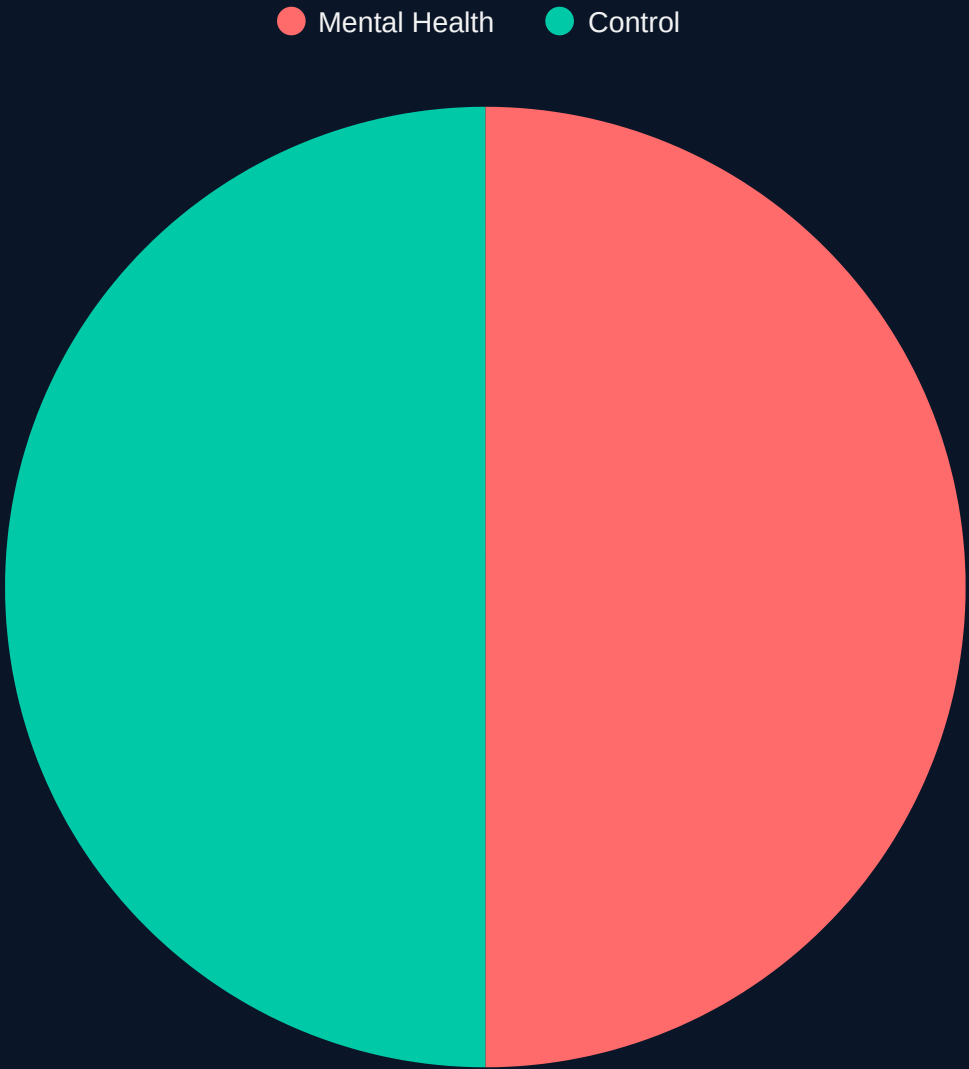
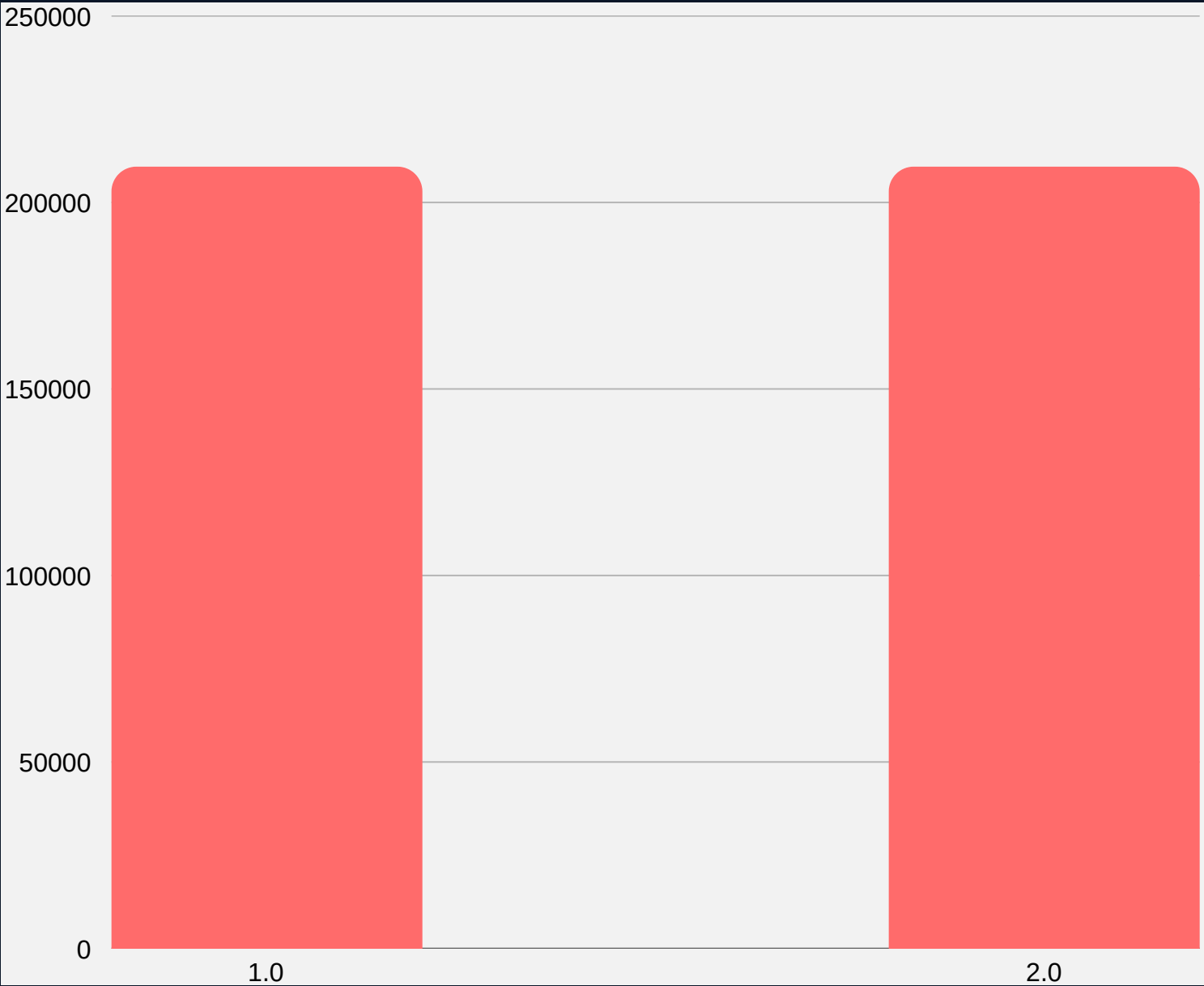
React Native · Docker · AWS

Mental Health Classification · Reddit Corpus



Results: Dataset Distribution

Balanced dataset of 419,180 posts · Mental Health vs. Control



Key Findings

- 209,590 samples per class — 50/50
- 17 mental health subreddits
- 11 control categories
- 17% control posts longer on avg
- 177.6 avg words in MH posts
- 153 avg words in control posts

Fig. 1 — Balanced Dataset: Mental Health (209,590) vs. Control (209,590) · Both distributions are right-skewed

Unique Vocabulary & Sentiment Analysis

Domain-specific terms and emotional polarity confirm strong class separability



-0.26

Mental Health Posts

Negative Bias

Avg Net Sentiment Score (VADER)

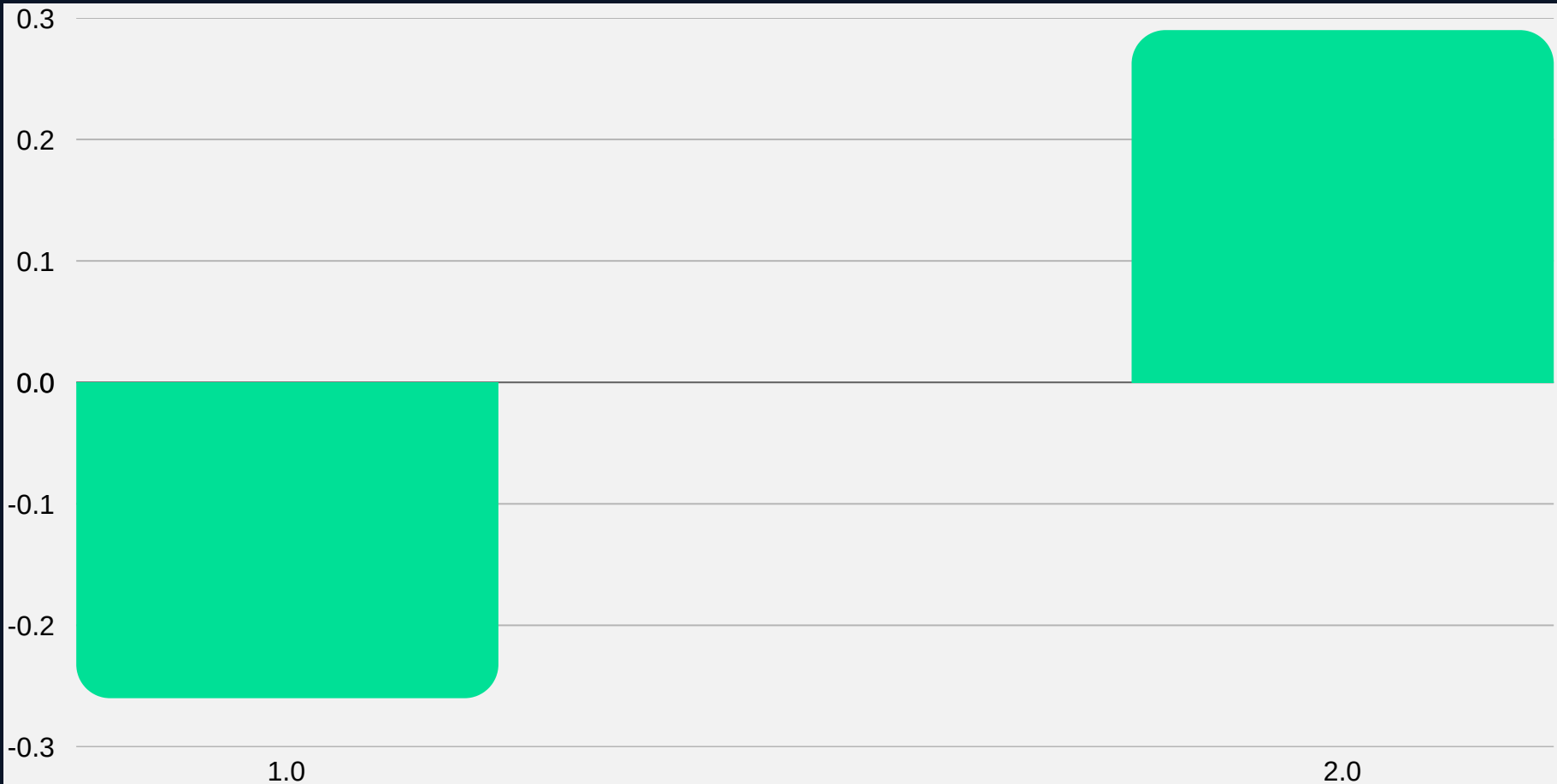


+0.29

Control Posts

Positive Bias

Avg Net Sentiment Score (VADER)



Key Observations

Sentiment gap of 0.55

strongly separates the two classes — a statistically significant signal.

MH posts contain

significantly more negative VADER-scored words per post.

Unique MH vocabulary

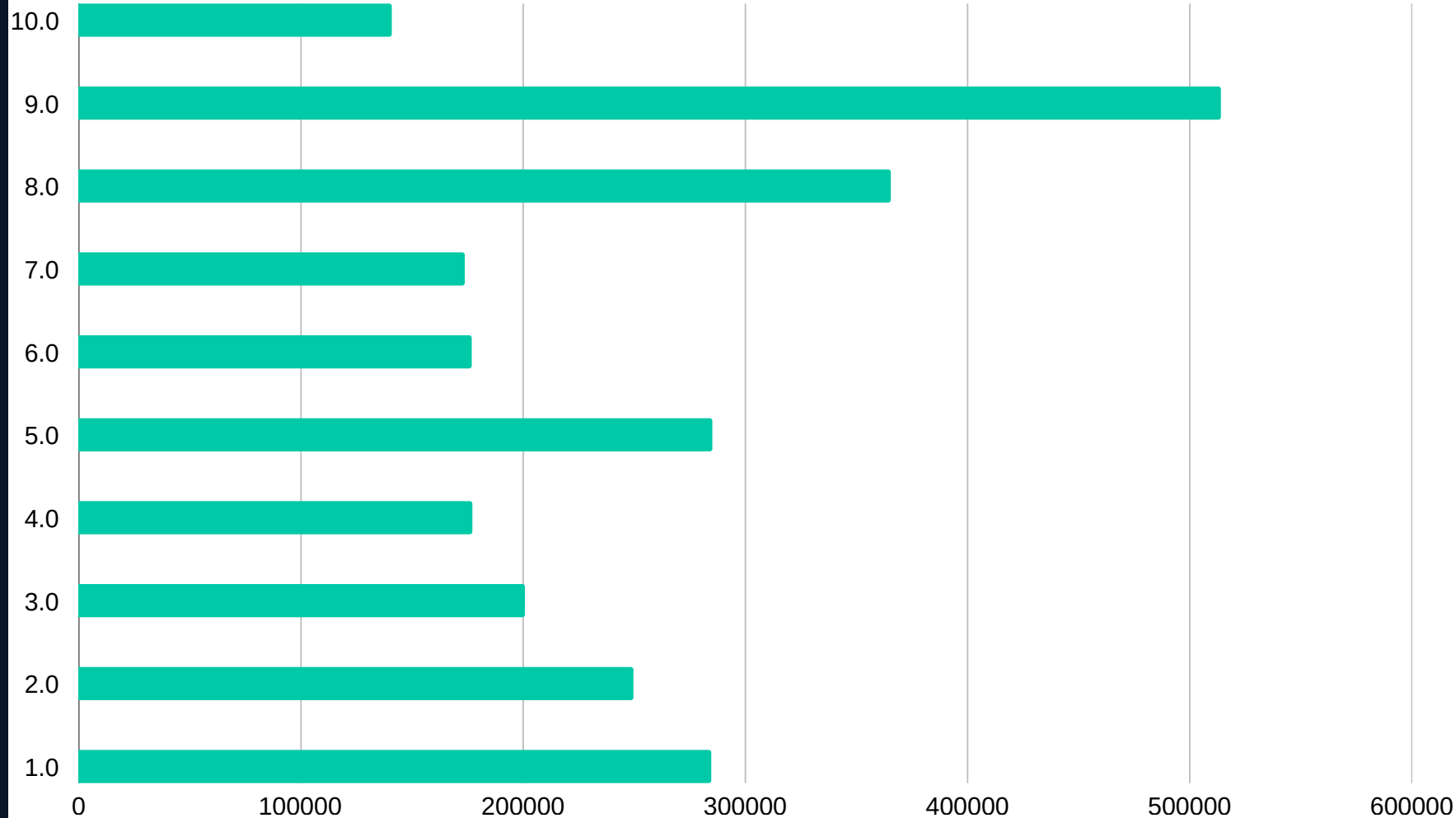
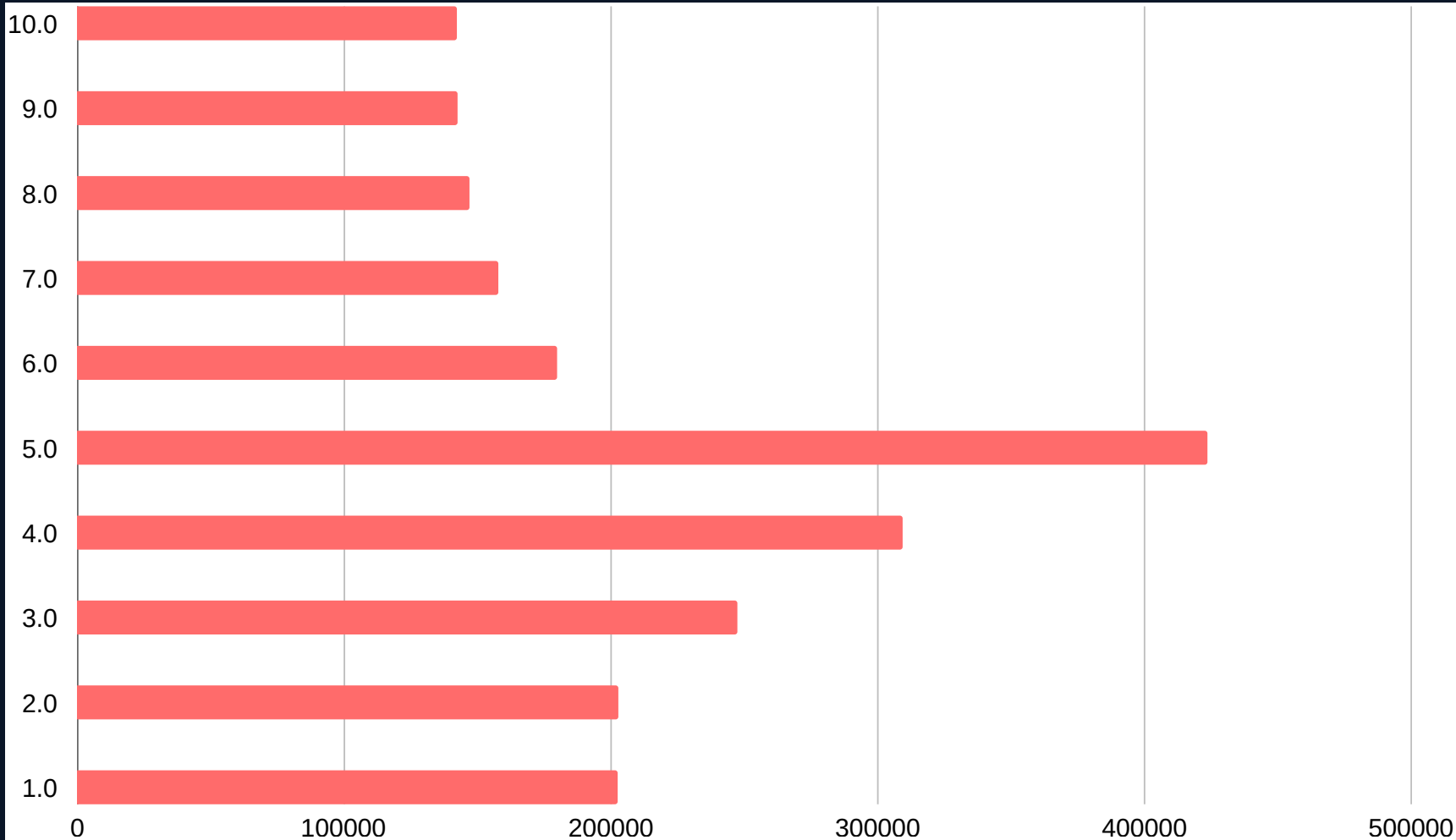
(medication names) further isolates the class from control.

Sentiment will be used

as a weighted feature in our downstream classification models.

Top Words & Unique Vocabulary

Frequency analysis uncovers discriminative vocabulary between classes



Unique MH Vocabulary (Medication Terms)

dexedrine · cymbalta · adhders · pdoc · strattera

mydayis · guanfacine · elvanse · depakote

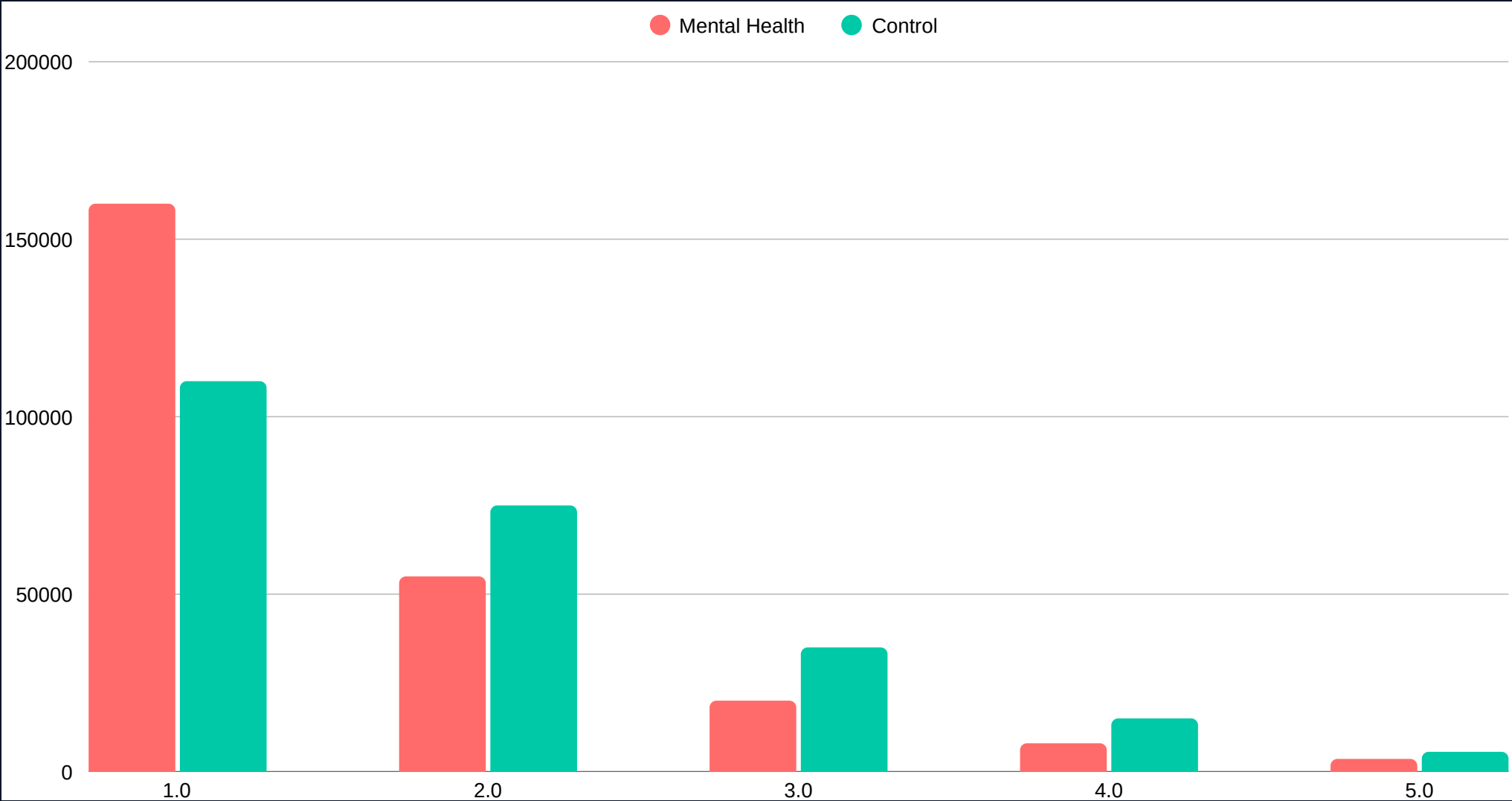
Unique Control Vocabulary (Financial Terms)

refinance · taxable · lender · schwab · jointly

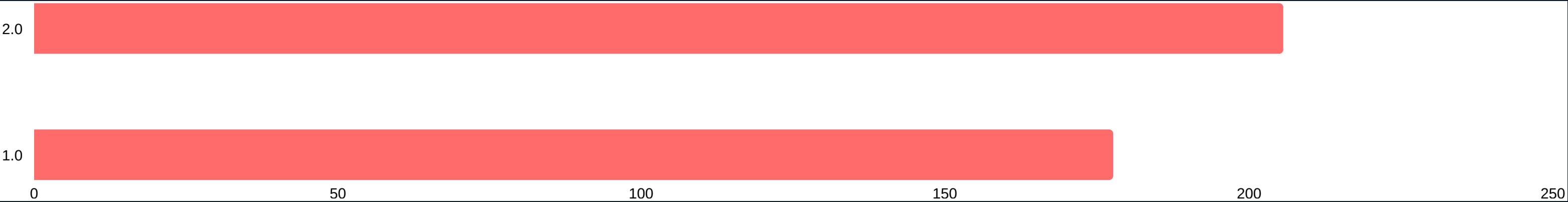
etfs · fco · iras · adjuster

Results: Text Length & Statistical Comparison

Control posts are significantly longer — reveals different writing patterns



Metric	Mental Health	Control	Δ Difference
Avg Length (chars)	934	1,092	+17% ↑
Median Length	665	812	+22% ↑
Std Deviation	944	1,025	+9% ↑
Avg Words	177.6	205.6	+16% ↑
Median Words	127	153	+20% ↑



Conclusion

Module 1 Complete



- 419,180-post Reddit dataset acquired & balanced
- Full preprocessing pipeline implemented
- EDA reveals clear statistical class differences
- Sentiment & vocabulary confirm separability

Key Insights



- MH posts: shorter, negative, medication-focused
- Control: longer, positive, domain-specific
- Net sentiment gap: -0.26 vs $+0.29$ ($\Delta 0.55$)
- Unique vocab confirms strong class separation

Foundation Established

- Cleaned, balanced dataset ready for training
- Statistical baselines set for comparison
- NLP features identified: length, sentiment, n-grams
- Next: SBERT classification - target $\geq 92\%$

Future Work — Phase 2 Roadmap

Planned directions: March – June 2026

M2	Semantic Classification  • Train BERT / RoBERTa for contextual text representation • Generate SBERT sentence embeddings for similarity analysis • Target: accuracy $\geq 92\%$, F1-score ≥ 0.88
M3	Temporal Pattern Analysis  • LSTM-based sequence model for emotional progression • Track mental health deterioration over 2–12 week windows • Detect gradual decline missed by static classifiers
M4	Explainable AI Integration  • Implement SHAP and LIME for prediction explainability • Highlight influential words and temporal patterns • Generate counsellor-readable risk explanations
SYS	Prototype & Validation  • Student web app + counsellor dashboard prototype • Multilingual NLP for Urdu-English code-switched text • Ethics review submission to FAST-NUCES IRB



Thank You!

Any Questions?

Stop Here!
Contains Previous Content

Ethical Considerations



Public and anonymized datasets only



No clinical diagnosis or medical claims



Research-focused academic use only

Conclusion

Empowering Students Through AI-Driven Mental Health Awareness

- ▶ Semantic & temporal analysis for deeper insights
- ▶ Early detection enables preventive support
- ▶ Explainable AI builds trust and understanding
- ▶ Student-focused approach addresses unique needs
- ▶ Research-oriented and academically rigorous
- ▶ High impact potential for student wellbeing

Feature Engineering

Text Analysis

- Suggest good exercises to manage anxiety
- Tell them they have anxiety/depression
- Provide mental health classification

Support Features

- Motivational speakers and counseling
- Personal AI chat assistant
- Text-based mental health insights

Analysis Methods

- Semantic analysis
- Temporal analysis
- Pattern recognition in text



Why Focus on Students?

1

High-Risk Group

Students face elevated risks for stress, anxiety, and burnout during critical developmental years

2

Data Gap

Existing datasets and systems focus on general users, missing unique academic contexts

3

Unique Patterns

Academic pressure and social stress create distinctive emotional patterns requiring specialized analysis

4

Prevention Potential

Early detection enables preventive interventions before issues escalate (non-clinical support)

Project Novelty

What Makes This Different

Unlike existing systems that classify individual posts, our project analyzes semantic meaning and emotional progression over time to assess student mental health trends with explainable predictions.

Temporal Analysis

Tracks emotional changes over time, not just snapshots

Student-Specific

Designed for academic stress patterns and student life contexts

Explainable AI

Shows which text signals triggered assessments

Multi-Model Ensemble

Combines NLP transformers with traditional ML for robustness

Comparison of Existing Mental Health Detection Systems vs MindCare

Study / Project	Source	Method Used	Dataset	Limitations	Link
Depression Detection using SBERT-CNN	ArXiv Research Paper	SBERT embeddings + CNN classifier	Reddit SMHD Dataset	Uses only social media text; no behavioural or psychological assessment integration	https://arxiv.org/abs/2302.02759
Deep Learning for Mental Illness Detection	Scientific Reports (Nature)	Deep Neural Network	Reddit mental health subreddits	Focuses only on disorder classification; lacks explainability and real system integration	https://www.nature.com/articles/s41598-020-684764-y
Transformer-Based Depression Detection	Springer / ACM Research	Transformer models (BERT / RoBERTa)	Reddit dataset	Binary classification only; does not provide multi-level risk scoring	https://link.springer.com/article/10.1007/s13278-024-01360-4
Stress Detection in Reddit Communities	NLP Research	Logistic Regression + Bag-of-Words	Dreaddit Dataset	Limited accuracy (~77%) and focuses only on stress detection	https://arxiv.org/abs/2312.01050
RSDD-Time Depression Dataset	Computational Social Science	Temporal dataset analysis	Reddit diagnosis posts	Provides dataset only; does not build a complete detection system	https://arxiv.org/abs/1806.07916
Mental Health Detection GitHub Project	GitHub Implementation	SVM / Logistic Regression	Small social media datasets	Experimental project; lacks dashboards, explainable AI, and real deployment	https://github.com/drmuskangarg/mentalhealthcare
MindCare (Our Project)	FYP System	SBERT + Sentiment Analysis + Multi-Modal Fusion Reddit dataset (~419k posts)		Comprehensive multi-modal assessment with psychological tests, behavioural analysis, explainable AI, and risk alert system	

Evaluation Metrics

Accuracy

$$(TP + TN) / Total$$

Overall correctness of predictions across all classes

Precision

$$TP / (TP + FP)$$

Accuracy of positive predictions (avoiding false alarms)

Recall

$$TP / (TP + FN)$$

Ability to find all relevant cases (sensitivity)

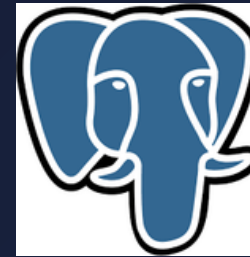
F1-Score

$$2 \times (Precision \times Recall) / (Precision + Recall)$$

Harmonic mean of precision and recall (balanced metric)

All metrics compared against SVM baseline performance

Tech Stack



Research Gap

1

Lack of semantic understanding using transformers

2

No temporal emotional modeling

3

Limited explainability in predictions

Proposed Solution



Transformer-based Semantic Analysis

BERT/SBERT for contextual understanding



Temporal Modeling

LSTM for tracking emotional trends



Explainable AI

SHAP/LIME for prediction transparency

System Capabilities & Limitations

DOES

- ✓ Analyze publicly available text
- ✓ Identify mental health indicators
- ✓ Track emotional trends over time
- ✓ Provide explainable predictions

DOES NOT

- ✗ Diagnose mental illness
- ✗ Monitor individuals without consent
- ✗ Access private data

This system supports awareness and early intervention, not clinical diagnosis

Model Comparison

SVM

✓ Simple implementation

✗ Low accuracy

LSTM

✓ Captures temporal patterns

✗ Weak semantic understanding

BERT

✓ Strong contextual semantics

✗ No time modeling

SBERT + LSTM

✓ Best balance: meaning + time

✗ Higher complexity

Problem: Not all students use social media, so additional data sources are required.

1. Psychological Questionnaires – PHQ-9 (Depression), GAD-7 (Anxiety), PSS (Stress).
2. AI Chatbot Interaction – Students talk with a mental-health chatbot; system analyzes sentiment and emotional keywords.
3. Behavioral Activity Tasks – Mood tracker, reaction-time tasks, journaling, and daily emotional check-ins.
4. Self-Reflection Journaling – Students write daily thoughts; NLP analyzes emotional patterns.
5. Multi-Modal Risk Scoring – Combines multiple inputs to generate final mental health risk level.



$$R = (0.4 \times A + (0.35 \times S) + (0.25 \times B))$$

where:

- A is psychological assessment score
- S is social media sentiment score
- B is behavioural activities score
- R is self-reflection journaling